CLASS 9 MATHEMATICS • CHAPTER 6

Measuring Space: Perimeter and Area

Exhaustive point-by-point presentation covering every concept, formula, historical derivation, and geometric theorem directly from the textbook.

✦ Akash Srivastva (www.akashmaths.online)

The Running Track: Why Are Starting Lines Staggered?

The Race Observation

At the start of a 4 × 100 m relay race:

- Athletes assemble in distinct lanes around the oval track.
- Runners in outer lanes start significantly ahead of inner lanes.
- Yet, all runners cross the exact same finish line!

What is a 'Stagger'?

The gap between adjacent starting marks:

- Outer circular curves have larger radii than inner curves.
- Without a stagger, outer-lane runners would cover a strictly longer path.
- Stagger precisely compensates for the extra perimeter of larger curves!

200 m vs 400 m School Track

Think and Reflect (Page 118):

- If a school builds a smaller 200 m track instead of a standard 400 m track, do we need a smaller stagger?
- To answer this mathematically, we must explore circular boundary lengths (circumference).

Classroom Check: The Track Stagger

1. Does the staggered starting line give runners in the outer lanes an unfair advantage in a 400 m race?

   Think about the total distance run by each athlete.

2. On which section of the running track do runners in all lanes cover the exact same distance: the straight stretches or the curved bends?

   Compare the geometry of straightaways vs semicircles.

3. Why do lanes on the outside curve have a larger perimeter than lanes on the inside curve?

   Consider how radius changes from inner to outer lanes.

4. What mathematical concept must the race organizers calculate to set the precise stagger distance between any two lanes?

   Focus on perimeter of circular arcs.

Perimeter as Boundary Walk & Invariant Ratios

DEFINITION OF PERIMETER

The perimeter of any closed plane shape is the total boundary length traversed by an insect walking once completely around its edge until it returns to the start.

Standard Polygons

- **Square (side a):** Perimeter = 4a units.
- **Equilateral △ (side a):** Perimeter = 3a units.
- **Rectangle (sides a, b):** Perimeter = 2(a + b) units. (Square is a special case where a = b).

Constant Perimeter Ratios

- When the side of a square doubles, its perimeter doubles too!
- **Square Ratio:** Perimeter : Side = **4 : 1** (strictly constant for all squares).
- **Equilateral △ Ratio:** Perimeter : Side = **3 : 1** (constant for all triangles).

The Circle Question

- What about a circle with radius r and diameter D?
- Does the ratio of Circumference (C) to Diameter (D) remain fixed regardless of circle size?
- Ancient thinkers discovered: **Yes, C / D is always constant!**

Classroom Check: Perimeter & Scaling

1 If the side of an equilateral triangle is tripled from 4 cm to 12 cm, what happens to the ratio of its perimeter to its side length?

Recall the invariant ratio property.

2 A rectangle has length 10 cm and width 5 cm. How can you show that the formula for a square's perimeter is a special case of the rectangle's perimeter?

Substitute equal dimensions into $2(a + b)$.

3 Does a circle have a fixed ratio between its boundary length and its width across the center? What is this ratio called?

Identify the C/D relationship.

4 If a circular coin and a giant Ferris wheel are compared, will the ratio of circumference to diameter be different?

Think about geometric similarity.

Estimating C/D: The Cotton Reel Experiment

Hands-On Home Experiment

A simple, highly accurate method to measure the C/D ratio at home:

- Take a cylindrical cotton reel and measure its diameter D accurately.
- Wrap a thin, strong thread tightly around the reel exactly **20 times**.
- Unwrap the thread, lay it straight, and measure total length L.
- Compute the ratio: **L / (20 × D)**.
- The result consistently lands between **3.1 and 3.2**!

The Constant Known as π

This universal constant bridges geometry and analysis:

- Named **π (pi)**, pronounced like 'pie' (not 'pizza').
- It represents the bridge connecting straight-edged polygons with nature's smooth curves.
- Can we compute π purely using geometric deduction, without physical measurements? *Ancient mathematicians answered yes!*

The Fundamental Circle Ratio: Circumference / Diameter = C / D = π (Constant for all circles)

Classroom Check: Measuring the C/D Ratio

1 Why is it much more accurate to wrap the thread 20 times rather than measuring just a single turn around the reel?

Consider how measurement errors divide over multiple turns.

2 Why does the experiment require the thread to be as thin as possible?

Think about how thread thickness affects the effective radius.

3 Between which two whole numbers does the C/D ratio always lie?

Recall the experimental bounds.

4 Can the value of π change if we move from measuring a micro-organism's cell cross-section to measuring the Earth's equator?

Think about scale-invariance.

Trapping π: From Mesopotamia to Archimedes

Mesopotamia (c. 1900 BCE)

Moving beyond crude integer 3:

- Observed that an inscribed regular hexagon in a unit circle (r = 1) has perimeter = 6.
- Diameter = 2 &implies; Hexagon perimeter / D = 3.
- Since circle curves outside hexagon: **π > 3**.
- Mesopotamians estimated: **π ≈ 3 + 1/8 = 3.125**.

Archimedes (250 BCE)

Pioneered rigorous geometric 'trapping':

- Inscribed and circumscribed regular polygons around a circle.
- Hexagon bounds: **3 < π < 2√3 ≈ 3.46**.
- Systematically doubled sides up to **96-sided polygons**!
- Established the famous inequality:
  3 (10/71) < π < 3 (1/7)
  (3.1408 < π < 3.1428)

Ptolemy of Alexandria (150 CE)

Astronomical precision:

- Refined Archimedes' computations for planetary chord tables.
- Derived sexagesimal fraction: **377 / 120 ≈ 3.14167**.
- Provided astronomy with unprecedented circular accuracy.

Classroom Check: Early History of π

1. Why does comparing a circle to an inscribed regular hexagon prove immediately that π must be strictly greater than 3?

   Calculate the perimeter of a regular hexagon inscribed in a circle of radius 1.

2. What was Archimedes' breakthrough method of "trapping" the value of π between two limits?

   Think about using both inner and outer polygons.

3. How many sides did the polygon have that Archimedes used to reach his celebrated bound 3 (10/71) < π < 3 (1/7)?

   Recall the polygon doubling count.

4. Which astronomer of Alexandria used the rational approximation 377 / 120 for planetary tables?

   Name the mathematician and time period.

Zu Chongzhi, Āryabhaṭa & Brahmagupta

Zu Chongzhi (480 CE)

The 24,576-polygon masterstroke:

- Carried Liu Hui's circle-cutting method to 24,576 sides!
- Found **Yuelü (Rough): 22/7 ≈ 3.1428**.
- Discovered **Milü (Close Ratio): 355/113 ≈ 3.1415929**.
- *World record for 800 years!* No fraction with denominator < 15,000 comes closer to π.

Āryabhaṭa (499 CE)

The concept of 'Āsanna' (Approaching):

- In Āryabhaṭīya, gave: **62832 / 20000 = 3.1416**.
- Crucially used the Sanskrit word **āsanna** (approximate/approaching).
- A profound intuition anticipating that π cannot be expressed as an exact finite fraction!

Brahmagupta (628 CE)

Elegance in equations:

- Suggested the value **π ≈ √10 ≈ 3.1622**.
- Chosen for algebraic convenience in quadratic calculations.
- Became the standard approximation in the medieval Arab world and Europe for centuries.

Classroom Check: Asian Mathematical Records

1 What was the remarkable fraction 355/113 discovered by Zu Chongzhi called, and why was it so significant?

Recall how many years it held the world accuracy record.

2 What philosophical and mathematical insight did Āryabhaṭa convey by calling his value 3.1416 "āsanna"?

Translate the meaning of the Sanskrit term.

3 Why did Brahmagupta suggest √10 for π despite it being slightly less accurate than Āryabhaṭa's value?

Think about ease of manipulation in algebraic equations.

4 How many sides did Zu Chongzhi calculate to establish his bounds for π?

State the immense polygon side count.

Madhava: First Exact Formula & Birth of Calculus

THE MADHAVA SERIES (C. 1380 CE)

$$\pi / 4 = 1 - 1/3 + 1/5 - 1/7 + 1/9 - 1/11 + \cdots$$

Paradigm Shift: From Geometry to Analysis

- **Madhava of Sangamagrāma (Kerala, India):** Realised π is not just an approximation to be trapped, but a *limit to be reached*.
- By replacing polygon-cutting with infinite numerical summation, Madhava birthed the foundational concepts of **Calculus** centuries before Europe!
- Calculated π accurately to **11 decimal places** (3.14159265358).

Subsequent Evolution & Trillions of Digits

- Kerala School successors: Nīlakaṇṭha (c. 1500) developed faster-converging series.
- John Machin (1706), Srinivasa Ramanujan (1914), and Chudnovsky brothers (1988) designed hyper-efficient algorithms.
- Today, π is known to over **100 trillion digits (neels)**!
- William Jones (1706) chose the symbol π for perimeter; Leonhard Euler made it universally famous.

Classroom Check: Madhava's Infinite Series

1. What fundamental conceptual leap did Madhava make that departed from all previous polygon-cutting methods?

   Consider the transition to infinite analytical series.

2. In Madhava's series for π / 4, what are the signs and denominators of the first four terms?

   Notice the alternating pattern and odd denominators.

3. Which Indian mathematical genius in 1914 formulated revolutionary, rapidly converging series for π that are still used by supercomputers today?

   Name the legendary 20th-century mathematician.

4. Why was the Greek letter π chosen to represent the circle constant by William Jones in 1706?

   Recall the Greek word for perimeter.

π is Irrational: Infinite, Non-Repeating Beauty

Cannot Be a Fraction

Rational numbers have repeating decimals:

- 1/3 = 0.333... ; 1/7 = 0.142857 142857...
- For π, digits go on forever with **zero repeating pattern**!
- π cannot be written as a/b (integers, b ≠ 0).
- Proven rigorously by **Lambert in 1761**.
- Therefore: **π ≈ 22/7, but π ≠ 22/7!**

Mnemonic Fun Fact

A fun way to remember digits of π:

- *"How I wish I could recollect pi."*
- Count letters in each word:
  How (3) • I (1) • wish (4) • I (1) • could (5) • recollect (9) • pi (2)
- Yields: **3.141592...**

Global Pi Celebrations

Annual dates honouring the constant:

- **Pi Day (March 14):** Written '3–14' in North America (3.14).
- **Pi Approximation Day (22 July):** Written '22/7' in India and Europe (22/7).

Classroom Check: Rational vs. Irrational

1 Is 22/7 a rational number or an irrational number? Is π a rational number or an irrational number?

Distinguish between the fraction and the real constant.

2 Why is it mathematically incorrect to write π = 22/7 instead of π ≈ 22/7?

Compare their decimal expansions.

3 Who was the first mathematician to rigorously prove that π is strictly irrational? In what year?

Recall the 18th-century European proof.

4 Why is July 22 celebrated as Pi Approximation Day in India and around the world?

Look at the calendar date format: Day / Month.

Arc Length Derived via Rotational Symmetry

Semicircle Arc (180°)

Full circle circumference = 2πr:

- A diameter splits the circle into 2 congruent semicircles.
- Reflection or 180° rotation swaps them identically.
- Length = 2πr ÷ 2 = **πr**.
- Formula view: **2πr × (180° / 360°)**.

Quarter Circle Arc (90°)

Rotational symmetry of 90°:

- Two perpendicular diameters divide the circle into 4 identical quadrants.
- Quarter-turn maps each arc to the next.
- Length = 2πr ÷ 4 = **πr / 2**.
- Formula view: **2πr × (90° / 360°)**.

General Arc Formula (θ°)

Direct proportionality to central angle:

- If an arc subtends central angle θ°:
- Fraction of the full circle = **θ° / 360°**.

Arc Length l = 2πr × (θ° / 360°)

Classroom Check: Arc Length Formulas

1 What is the length of an arc that subtends a central angle of 60° in a circle of radius 21 cm? (Use $\pi \approx 22/7$)

Compute $(60/360) \times 2 \times (22/7) \times 21$.

2 If the central angle of an arc is doubled while keeping the radius fixed, what happens to the length of the arc?

Look at the direct proportionality to θ.

3 What is the total perimeter of a semicircular protractor of radius r? (Careful: curved edge + straight diameter!)

Sum the arc length and the diameter: $\pi r + 2r$.

4 What fraction of the total circumference is cut off by a central angle of 45°?

Calculate $45° / 360°$.

Inside the Geometry of an Olympic 400 m Track

The Exact Dimensions & Total Distance

Standard International Athletics track layout:

- **2 Straight Sections:** Each measures **84.39 m**.
  Total straight distance = 2 × 84.39 = **168.78 m**.
- **2 Semicircular Bends:** Inner curb radius = **36.5 m**.
- Athletes run 0.3 m away from the curb &implies; running radius = **36.8 m**.
- Both semicircles form one full circle:
  Circumference = 2 × π × 36.8 m = **231.22 m**.
- **Total Circuit:** 168.78 m + 231.22 m = **400.00 m**!

↔ Computing the Lane Stagger

Why Lane 2 must start ahead:

- Standard lane width = **1.22 m**.
- On straight stretches, runners in all lanes cover identical distance.
- In Lane 2, radius increases by 1.22 m!
- Extra curve distance in one complete lap:
  $2\pi(r + 1.22) - 2\pi r = \mathbf{2\pi \times 1.22\ m \approx 7.67\ m}$!
- Therefore, Lane 2 must start **7.67 m ahead** of Lane 1 to run the exact same 400 m race!

Classroom Check: Track Geometry

1 Why is an athlete's running radius taken as 36.8 m rather than the curb radius of 36.5 m?

Consider the distance the runner stays away from the inner curb line.

2 If the lane width is 1.22 m, will the stagger needed between Lane 2 and Lane 3 be the same as the stagger between Lane 1 and Lane 2?

Check whether the difference in radius between adjacent lanes is constant.

3 In a 100 m sprint run entirely on the straightaway, is any lane stagger needed? Why or why not?

Recall where the extra distance comes from.

4 Does the stagger formula $2\pi w$ depend on the radius of the track, or purely on the lane width w?

Evaluate $2\pi(R_2 - R_1)$.

Perimeter Paradoxes: Equal Paths & Overlaps

Puzzle 1: Circles Through Centers (Ex 1)

Two circles of radius r pass through each other's center:

- △ABC is equilateral (AB = AC = BC = r) &implies; ∠CAB = 60°.
- By symmetry, central angle ∠CAD = 60° + 60° = 120°.
- Each dotted inner arc subtends 120° (which is 1/3 of circumference).
- Outer solid arc = 1 – 1/3 = **2/3 of circle**.
- Total Perimeter of both outer arcs:
  $2 \times (2/3 \times 2\pi r) = \mathbf{(8/3)\pi r}$!

Puzzle 2: The Semicircular Paths Paradox (Ex 2)

Path A (one giant semicircle) vs Path B (three small semicircles):

- Let the diameter PQ = 2a'. Semicircle a has length = **πa'**.
- Path b + c + d has diameters 2b' + 2c' + 2d' = 2a'.
- Lengths of small arcs: πb' + πc' + πd' = **π(b' + c' + d')**.
- Since b' + c' + d' = a', **Length of Path A = Length of Path B!**
- *A stunning counter-intuitive result: Both paths are exactly equal!*

Classroom Check: Perimeter Puzzles

1 In the Semicircular Paths Paradox, if Path B were made of 100 tiny semicircles instead of 3, would its total length still equal Path A?

Check if the sum of diameters always equals PQ.

2 In Example 1, why is △ABC guaranteed to be an equilateral triangle?

List the lengths of sides AB, BC, and CA in terms of radius r.

3 What is the perimeter of a circular flower with 4 semicircular petals constructed on the sides of a square of side 10 cm?

Notice that 4 semicircles equal 2 complete circles!

4 If a bicycle wheel of diameter 56 cm rotates 1000 times, how many meters has the cyclist traveled?

Calculate $1000 \times (\pi \times D)$.

Area of Parallelograms & The Thin Shear

Area of a Rectangle (A = ab)

- Area is the 2D space occupied on a plane, measured in unit squares (1 unit × 1 unit = 1 sq. unit).
- A grid of a rows and b columns yields: **Area = a × b = ab sq. units**.
- Area of square of side a is the special case: **$A = a^2$**.

Transforming Parallelogram to Rectangle

- Cutting a right triangle from one side of a parallelogram and pasting it to the opposite side creates a rectangle!
- Same base b, same perpendicular height h.
- **Area of Parallelogram = Base × Height = bh**.
- *Think and Reflect:* Can we find a parallelogram's area from side lengths alone? **No!** Varying the angle changes height h and alters area without changing side lengths.

THE "THIN PARALLELOGRAM" GAP & CAVALIERI FIX

If a parallelogram is so slanted that the altitude drops outside the base, shift the top side by congruent increments ($\triangle CDD' \cong \triangle BAA'$) repeatedly until an altitude falls within the base! Area remains strictly invariant.

Classroom Check: Parallelogram Geometry

1 Why can two different rhombuses have side lengths of 5 cm each, but completely different areas?

Think about how changing the internal corner angles affects height.

2 In the transformation of a parallelogram into a rectangle, what two dimensions remain strictly identical?

Identify the base and the perpendicular height.

3 Why does the standard single-triangle cut fail when a parallelogram is extremely slanted ("thin")?

Consider where the foot of the altitude falls.

4 What is the area of a parallelogram with base 14 cm and perpendicular height 7.5 cm?

Calculate base × height.

Area of a Triangle & The Median Theorem

Two Triangles Make a Parallelogram

- Any two congruent copies of △ABC fit together along a shared side to form a parallelogram ABCD!
- Area of the parallelogram = base × height = bh.
- Therefore: **Area of Triangle = ½ bh**.
- Valid for acute, right, and obtuse triangles.

Polygon Scissors Congruence

Think and Reflect (Wallace-Bolyai-Gerwien):

- Any two polygons P and Q with equal area can always be cut into a finite number of polygonal pieces and reassembled to form each other!
- Demonstrates deep geometric conservation of area under dissection.

KEY THEOREM: MEDIAN DIVIDES TRIANGLE INTO EQUAL AREAS

A median of a triangle divides it into two triangles of equal area!

Proof: Let AD be median of △ABC. △ABD and △ACD have equal bases (BD = DC) and share the exact same perpendicular height h. Area(△ABD) = ½(BD)(h) = ½(DC)(h) = Area(△ACD). (Even though they are NOT congruent!)

Classroom Check: Triangles & Medians

1 If two triangles have identical areas, must they always be congruent to each other?

Think about the two triangles created by a median.

2 Why do △ABD and △ACD have the exact same height h when AD is a median from vertex A?

Look at the common vertex A and the shared base line BC.

3 If the three medians of a triangle intersect at the centroid G, into how many triangles of equal area is the triangle divided?

Recall the property of the centroid dividing the area into 6 equal parts.

4 Among all rectangles with a fixed perimeter of 40 units, which shape has the maximum area? What are its dimensions?

Test dimensions 10 × 10 vs 15 × 5 vs 19 × 1.

Heron’s Formula: Area from Three Sides Alone

HERON'S MASTER FORMULA (C. 60 CE, ALEXANDRIA)

Semi-perimeter: s = ½(a + b + c) &implies; Area = √[s(s – a)(s – b)(s – c)]

Equilateral △ (Side a)

- s = 3a / 2
- s – a = a / 2 (for all 3 sides)
- Area = √[(3a/2)(a/2)³] = **(√3 / 4) a²**
- Matches standard ½ × base × (√3/2 a) height perfectly!

Isosceles △ (Sides a, a, 2b)

- s = ½(2a + 2b) = a + b
- s – a = b, s – 2b = a – b
- Area = √[(a + b)(b)(b)(a – b)] = **b√(a² – b²)**
- Matches Baudhāyana-Pythagoras height derivation!

Right Triangle (3, 4, 5)

- s = ½(3 + 4 + 5) = 6
- s – a = 3, s – b = 2, s – c = 1
- Area = √[6 × 3 × 2 × 1] = √36 = **6 sq. units**
- Matches ½ × 3 × 4 = 6 sq. units without needing height!

Classroom Check: Heron’s Formula

1 What is the semi-perimeter s of a triangle with side lengths 13 cm, 14 cm, and 15 cm?

Calculate ½(13 + 14 + 15).

2 Why is Heron's formula considered vastly superior to ½ bh when dealing with land surveys and irregular terrain?

Think about the difficulty of measuring internal perpendicular altitudes in the field.

3 If the side lengths of an equilateral triangle are 6 cm each, what is its area?

Apply (√3 / 4) a^2.

4 Can Heron's formula be applied if s – a turns out to be zero or negative? What does that tell you about the triangle?

Recall the Triangle Inequality theorem.

Triangle Area Linked to Incircle & Circumcircle

◎ Incircle Formula: Area = r · s

The circle that fits tightly inside the triangle, touching all 3 sides:

- Let inradius = r. Join incenter to the 3 vertices to form 3 smaller triangles.
- Each small triangle has altitude r and base a, b, or c.
- Total Area = ½ar + ½br + ½cr = **r(a + b + c) / 2 = r · s**.
- Allows instant calculation of inradius: **r = Area / s**!

● Circumcircle Formula: Area = abc / 4R

The circle passing through all 3 vertices A, B, and C:

- Let circumradius = R.
- Area relates directly to the product of all three sides:
  Area of △ABC = abc / (4R)
- Allows instant calculation of circumradius:
  R = abc / (4 × Area)!
- Both formulas demonstrate breathtaking geometric symmetry.

Classroom Check: Inradius & Circumradius

1 For a right triangle with sides 3, 4, 5 and area 6, what is its semi-perimeter s, and what is its inradius r?

Use r = Area / s = 6 / 6.

2 For the same 3–4–5 triangle, what is its circumradius R using abc / (4 × Area)?

Compute (3 × 4 × 5) / (4 × 6).

3 Where does the center of the circumcircle lie for any right-angled triangle?

Connect this to R = hypotenuse / 2 = 5 / 2 = 2.5!

4 How many circles can touch all three sides of a given triangle from the inside?

Recall the uniqueness of the incircle.

Brahmagupta's Formula for Cyclic Quadrilaterals

BRAHMAGUPTA'S FORMULA (628 CE, BRAHMASPHUTASIDDHANTA)

For a cyclic quadrilateral with sides a, b, c, d and semi-perimeter $s = \frac{1}{2}(a + b + c + d)$:

$$\text{Area} = \sqrt{(s - a)(s - b)(s - c)(s - d)}$$

Why 4 Sides Alone Are Not Enough

- A 4–gon with sides 3, 3, 3, 3 can be flexed into a square (Area = 9) or squashed rhombuses (Area = 8.01, 5.41...).
- Four side lengths alone CANNOT fix the area of a general quadrilateral.
- **Cyclic condition fixes the angles rigidly**, enabling an exact area formula!

Verification on Special Cases

- **Rectangle (sides a, b, a, b):** All rectangles are cyclic. $s = a + b$.
  Area $= \sqrt{[b \cdot a \cdot b \cdot a]} = \mathbf{ab}$ (Proven!).
- **Isosceles Trapezium:** All are cyclic.
  Formula simplifies to $\mathbf{(a + b)h}$, matching standard trapezium area!

⚡ QUICK BOX 15

CLASSROOM Q&A LIVE

Classroom Check: Brahmagupta’s Formula

1. Can Brahmagupta's formula be used to find the area of ANY quadrilateral, or only cyclic quadrilaterals?

   Recall why non-cyclic quadrilaterals can flex into different areas.

2. Why is an isosceles trapezium always guaranteed to be a cyclic quadrilateral?

   Check whether its opposite angles add up to 180°.

3. If a cyclic quadrilateral has side lengths 5, 5, 12, 12, what is its semi-perimeter s?

   Calculate ½(5 + 5 + 12 + 12).

4. In which historical century and treatise did Brahmagupta publish this crowning achievement of ancient geometry?

   Name the century (7th century CE) and Indian text.

Brahmagupta's Formula Generalises Heron's Formula

The Concept of Generalisation

- A **special case** results when an extra condition is applied to a general theorem.
- A square is a special case of a rectangle (where b = a).
- An isosceles right triangle is a special case of the Pythagorean theorem: $a^2 + a^2 = c^2$ &implies; $c = a\sqrt{2}$.
- Algebraic identities: $(a + b)^2$ is a special case of $(a + b + c)^2$ when c = 0!

Setting Side d = 0

Treating a triangle as a 4–gon with one side collapsed to length zero:

- Let the 4th vertex coincide with the 1st vertex (d = 0).
- Every triangle is cyclic (unique circumcircle exists!).
- Semi-perimeter: $s = \frac{1}{2}(a + b + c + 0) = \frac{1}{2}(a + b + c)$.
- Substitute d = 0 into Brahmagupta:
  Area = $\sqrt{[(s - a)(s - b)(s - c)(s - 0)]}$
  = $\sqrt{\mathbf{[s(s - a)(s - b)(s - c)]}}$!
- **Heron's formula is a direct special case of Brahmagupta!**

Classroom Check: Generalisation Principles

1 What geometric condition transforms Brahmagupta's 4-gon formula directly into Heron's triangle formula?

Set the length of the fourth side d to 0.

2 Why are we guaranteed that a triangle with d = 0 satisfies the cyclic condition required by Brahmagupta?

Recall Theorem 1 from Chapter 5 (Circumcircle of 3 points).

3 Show how the perimeter of a square (4a) is a special case of the perimeter of a rectangle (2(a + b)).

Substitute b = a.

4 In algebraic identities, how is $(a + b)^2$ a special case of $(a + b + c)^2$?

Substitute c = 0 into the trinomial expansion.

Squaring a Rectangle: Baudhāyana's Śhulbasūtra

THE ANCIENT CHALLENGE (800 BCE)

To "square a shape" means to construct a square having the exact same area as that shape. Baudhāyana demonstrated how to construct a square equal in area to a rectangle of sides a and b (where a > b).

The Geometric Construction Steps

- Given rectangle ABCD with length a and breadth b.
- Cut off square of side b; find midpoint F of remaining segment.
- Draw square of side AF = (a + b) / 2.
- Sweep circular arc of radius (a + b) / 2 to cut rectangle edge at K.
- Construct square on HP. Square HPQS has area = **ab**!

The Algebraic Translation

Baudhāyana translated the algebraic identity into geometry:

$$[(a + b)/2]^2 - [(a - b)/2]^2 = ab$$

- Hypotenuse = HK = (a + b)/2
- Base = BH = (a – b)/2
- By Baudhāyana-Pythagoras theorem:
  $HP^2 = HK^2 – BH^2 = ab$!

Classroom Check: Baudhāyana's Construction

1 What did the phrase "to square a shape" mean in ancient Indian and Greek mathematics?

Define the construction objective.

2 Which famous algebraic identity represents the exact mathematical foundation of Baudhāyana's geometric squaring method?

State the difference of squares of average and half-difference.

3 If a rectangle has sides 8 cm and 2 cm (Area = 16 cm^2), what is the side length of the square constructed by Baudhāyana's method?

Calculate √16.

4 How could you use Baudhāyana's squaring technique to square a given triangle?

Hint: First transform the triangle into a rectangle of equal area!

Area of a Circle: From C^2/A Ratios to πr^2

Perimeter² : Area Constant

For any shape, P^2 / Area is constant:

- Square: $(4a)^2 / a^2 =$ **16 : 1**.
- Equilateral △: $(3a)^2 / [(\sqrt{3}/4)a^2] =$ **36 : √3**.
- Therefore, for a circle, C^2 / Area must also be a constant!
- Babylonians (1500 BCE): $C^2 / A \approx 12$ &implies; **$A \approx C^2 / 12$**.

Ancient Egyptian & Vedic Formula

Independent discovery of $(256/81)r^2$:

- Egyptians (1500 BCE): Area $\approx (8d / 9)^2$.
- Substitute $d = 2r$:
  Area $\approx (256 / 81)\, r^2 \approx 3.1605\, r^2$.
- Amazingly, the *exact same formula* appears in the **Baudhāyana Śhulbasūtra (800 BCE)**!

Archimedes' Theorem (250 BCE)

Unifying circumference and area with π:

- Proved: Area of circle = Area of right triangle with legs equal to radius r and circumference C!
- Area = ½ × Circumference × Radius
- **Area = ½ × (2πr) × r = πr^2**!

Classroom Check: History of Circle Area

1 What constant ratio does the square of the perimeter divided by the area (P^2 / A) yield for all squares?

Calculate $(4a)^2 / a^2$.

2 Which ancient civilisation used the formula $A \approx C^2 / 12$ to calculate the area of circular grain silos?

Name the Mesopotamian civilisation.

3 What decimal value does the ancient Egyptian and Baudhāyana formula (256 / 81) give for π?

Divide 256 by 81.

4 How did Archimedes express the area of a circle in terms of a right-angled triangle?

State the two perpendicular leg lengths of the triangle.

Nīlakaṇṭha's Slice Proof & Sector Formulas

Nīlakaṇṭha Somayājī's Slice Proof (c. 1500)

The most intuitive visual proof of $A = \pi r^2$:

- Slice a circular disc into many thin sectors and interlock them alternately into a strip.
- As slices get narrower, curved edges straighten into lines.
- Figure forms a parallelogram with:
  - **Base** = half circumference = $\frac{1}{2}(2\pi r) = \pi r$.
  - **Height** = radius = r.
- **Area of Circle = Base × Height = $(\pi r) \times r = \pi r^2$!**

Sector & Segment of a Circle

Proportional area by central angle:

- **Semicircular Disc (180°):** $\frac{1}{2}\pi r^2$.
- **Quarter Circular Disc (90°):** $\frac{1}{4}\pi r^2$.
- **General Sector (subtends θ°):**

$$\text{Area of Sector} = \pi r^2 \times (\theta^\circ / 360^\circ)$$

- **Segment Area:** Area of sector – Area of triangle!

Classroom Check: Slices & Sectors

1 In Nīlakaṇṭha's slice rearrangement proof, why is the base of the resulting parallelogram equal to πr?

Think about which half of the circumference forms the top and bottom edges.

2 What is the area of a sector of a circle with radius 7 cm and central angle 60°?

Compute (60/360) × (22/7) × 7^2.

3 What is the difference between a "sector" of a circle and a "segment" of a circle?

Distinguish between boundaries formed by radii vs chords.

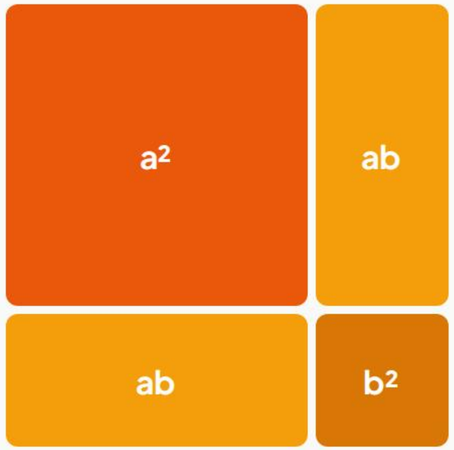
4 If the minute hand of a clock is 7 cm long, what area does it sweep in 10 minutes?

Determine the central angle swept in 10 minutes (60°).

Visual Proofs: Algebraic Identities as Area Models

Area Model of $(a + b)^2$

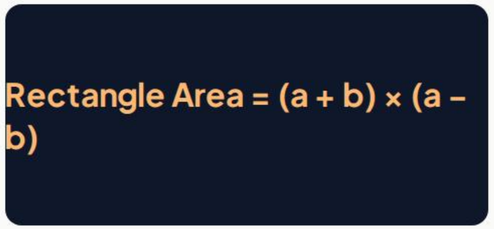
Large square of side $(a + b)$ partitioned into 4 regions:



$$(a + b)^2 = a^2 + 2ab + b^2$$

Area Model of $a^2 - b^2 = (a + b)(a - b)$

Square of side a with a $b \times b$ corner removed and rearranged:



Cut the L-shaped region of area $(a^2 - b^2)$ and reassemble into a single rectangle!

$$a^2 - b^2 = (a + b)(a - b)$$

Classroom Check: Geometric Area Models

1 How many total sub-regions are formed when you construct the geometric area model for $(a + b + c)^2$?

Count the squares and cross-product rectangles: 3 squares + 6 rectangles = 9 regions!

2 In the geometric rearrangement for $a^2 - b^2$, what are the length and width of the newly assembled rectangle?

Identify the factors $(a + b)$ and $(a - b)$.

3 Why is geometric area modeling a powerful pedagogical tool for understanding algebraic identities?

Connect visual physical areas to abstract symbolic formulas.

4 How can an $(a - b)^2$ square be modeled by subtracting areas from a master square of side a?

Recall $a^2 - 2ab + b^2$.

Model Problem Solutions from Exercises

Problem 1: Triangle in Rectangle (Ex 6.2 Q1)

Given: Rectangle ABCD of 10 cm × 8 cm. E on BC. Find Area(△ADE).

- Base AD = 10 cm.
- Perpendicular height from E to AD equals AB = 8 cm.
- Area = ½ × Base × Height = ½ × 10 × 8 = **40 cm²**!
- *Notice: Position of E on BC does NOT change the area!*

Problem 2: Concentric Ring Chord (Ex Q23)

Given: Chord BC of length l is tangent to inner circle at A.

- Let outer radius = R, inner radius = r.
- A bisects BC &implies; AB = l / 2.
- Right △OAB: $R^2 - r^2 = (l/2)^2 = l^2/4$.
- Ring Area = $\pi R^2 - \pi r^2 = \pi(R^2 - r^2) = \mathbf{\frac{1}{4}\pi l^2}$!

Problem 3: Lunes of Hippocrates (Ex Q24)

Given: Semicircles on sides of right △ with legs a, b and hypotenuse c.

- Area(A) + Area(B) = $(\pi/8)(a^2 + b^2)$ – (Segment portions).
- By Pythagorean theorem, $a^2 + b^2 = c^2$.
- Algebraic cancellation reveals: **Area(Lune A) + Area(Lune B) = Area(Triangle C)**!

Master Perimeter & Area Reference Checklist

Boundary & Circumference Principles

- ✅ **Perimeter:** Total boundary walk distance. Invariant scaling ratios (Square 4:1, Equilateral △ 3:1).
- ✅ **Circle Ratio:** $C / D = \pi \approx 22/7 \approx 3.1416$. Circumference $C = 2\pi r = \pi D$.
- ✅ **Arc Length:** $l = 2\pi r \times (\theta° / 360°)$. Semicircle = πr; Quadrant = $\pi r/2$.
- ✅ **Historical Estimates:** Zu Chongzhi (355/113), Āryabhaṭa (3.1416 āsanna), Brahmagupta ($\sqrt{10}$).
- ✅ **Calculus Birth:** Madhava's infinite series: $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$ (11 digits).
- ✅ **Irrationality:** Lambert (1761) proved π is irrational.

2D Area & Master Formulas

- ✅ **Parallelogram & Triangle:** Parallelogram = bh; Triangle = $\frac{1}{2}bh$.
- ✅ **Median Theorem:** A median divides any triangle into two equal-area triangles.
- ✅ **Heron's Formula:** Area = $\sqrt{[s(s-a)(s-b)(s-c)]}$, where $s = \frac{1}{2}(a+b+c)$.
- ✅ **Inradius & Circumradius:** Area = $r \cdot s = abc / (4R)$.
- ✅ **Brahmagupta's Formula:** Cyclic 4–gon Area = $\sqrt{[(s-a)(s-b)(s-c)(s-d)]}$. Generalises Heron when $d = 0$!
- ✅ **Baudhāyana's Squaring:** $[(a+b)/2]^2 - [(a-b)/2]^2 = ab$.
- ✅ **Circle & Sector Area:** Area = πr^2; Sector = $\pi r^2 \times (\theta° / 360°)$.

END OF CHAPTER 6

THANK YOU

"Mathematics is the queen of sciences, and the geometry of space is its crown — connecting the straight lines of polygons to the infinite curves of nature."

✦ Akash Srivastva (www.akashmaths.online)